$\therefore \quad A \cap B = \{(4, 5, 6), (4, 6, 5)\}$

$\therefore \quad P(A/B) = \dfrac{2}{36} = \dfrac{1}{18}$

ILLUSTRATION 14.6

Assume that each born child is equally likely to be a boy or a girl. If a family has two children, what is the conditional probability that both are girls given that

(i) the youngest is a girl, **(ii)** at least one is a girl?

Sol. Let b and g represent the boy and the girl child, respectively. If a family has two children, the sample space will be

$S = \{(b, b), (b, g), (g, b), (g, g)\}$

Let A be the event that both children are girls. Therefore,

$A = \{(g, g)\}$

(i) Let B be the event that the youngest child is a girl. Therefore,

$B = \{(b, g), (g, g)\}$

$\Rightarrow A \cap B = \{(g, g)\}$

$\therefore \quad P(B) = \dfrac{2}{4} = \dfrac{1}{2}$ and $P(A \cap B) = \dfrac{1}{4}$

The conditional probability that both are girls, given that the youngest child is a girl, is given by $P(A/B)$.

$$\therefore \quad P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(ii) Let C be the event that at least one child is a girl. Therefore,

$C = \{(b, g), (g, b), (g, g)\}$

$\Rightarrow A \cap C = \{g, g\}$

$\Rightarrow P(C) = \dfrac{3}{4}$ and $P(A \cap C) = \dfrac{1}{4}$

The conditional probability that both are girls, given that at least one child is a girl, is given by $P(A \mid C)$. Therefore,

$$P(A/C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}$$

ILLUSTRATION 14.7

Consider the experiment of tossing a coin. If the coin shows head, toss it again but if it shows tail, then throw a dice. Find the conditional probability of the event 'the dice shows a number greater than 4' given that there is at least one tail.

Sol. According to the question,

$S = \{(H, H), (H, T), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$

In the above set, there are 8 elementary events, but all are not equally likely.

However, events (H, H) and (H, T) are equally likely. Each event has probability $\dfrac{1}{4}$.

Events $(T, 1), (T, 2), \ldots, (T, 6)$ are equally likely and each event has probability $\dfrac{1}{12}$.

Let event A be "the dice shows a number greater than 4"

And event B be "there is at least one tail".

$\therefore \quad A = \{(T, 5), (T, 6)\}$

and $\quad B = \{(H, T), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$

$\therefore \quad A \cap B = \{(T, 5), (T, 6)\}$

$\therefore$ Required probability, $P(A/B) = \dfrac{P(A \cap B)}{P(B)}$

$$= \frac{\frac{1}{12} + \frac{1}{12}}{\frac{1}{4} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}} = \frac{1/6}{3/4} = \frac{2}{9}$$

ILLUSTRATION 14.8

A box contains 10 mangoes out of which 4 are rotten. Two mangoes are taken out together. If one of them is found to be good, then find the probability that the other is also good.

Sol. Let A be the event that the first mango is good, and B be the event that the second one is good. Then, required probability is

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

Now, probability that both mangoes are good is

$$P(A \cap B) = \frac{{}^6C_2}{{}^{10}C_2}$$

Probability that first mango is good is

$$P(A) = \frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^6C_1 \times {}^4C_1}{{}^{10}C_2}.$$

Hence, $P(B/A) = \dfrac{{}^6C_2}{{}^6C_2 + {}^6C_1 \times {}^4C_1} = \dfrac{15}{15+24} = \dfrac{5}{13}$

CONCEPT APPLICATION EXERCISE 14.1

1. Three coins are tossed. If one of them shows tail, then find the probability that all three coins show tail.
2. If two events A and B are such that $P(A') = 0.3$, $P(B) = 0.4$ and $P(A \cap B') = 0.5$, then find the value of $P[B/(A \cup B')]$.
3. In a single throw of two dice what is the probability of obtaining a number greater 7, if 4 appears on the first dice?
4. A coin is tossed three times in succession. If E is the event that there are at least two heads and F is the event in which first throw is a head, then find $P(E/F)$.
5. Consider a sample space "S" representing the adults in a small town who have completed the requirements for a collage degree. They have been categorized according to sex and employment as follows:

	Employed	Unemployed
Male	460	40
Female	140	260

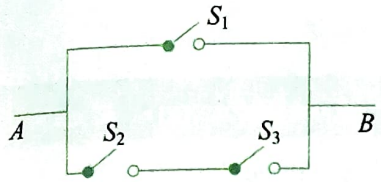
An employed person is selected at random. Find the probability that the chosen one is a male.

ANSWERS

1. 1/7 **2.** 1/4 **3.** 1/2 **4.** 3/4 **5.** 23/30

ILLUSTRATION 14.18

An electrical system has open-closed switches S_1, S_2 and S_3 as shown.



The switches operate independently of one another and the current will flow from A to B either if S_1 is closed or if both S_2 and S_3 are closed.

If $P(S_1) = P(S_2) = P(S_3) = \frac{1}{2}$, then find the probability that the circuit will work.

Sol. $P(S_1) = P(S_2) = P(S_3) = \frac{1}{2}$

Let E be event that "the current will flow".

$$P(E) = P((S_2 \cap S_3) \cup S_1))$$
$$= P(S_2 \cap S_3) + P(S_3) - P(S_1 \cap S_2 \cap S_3)$$
$$= P(S_2)\,P(S_3) + P(S_3) - P(S_1)P(S_2)P(S_3)$$
$$= \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2} - \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{5}{8}$$

ILLUSTRATION 14.19

The odds against a certain event are 5:2 and the odds in favor of other event, independent of the former, are 6:5. Then find the probability that at least one of the events will happen.

Sol. $P\{\text{First event does not happen}\} = \frac{5}{5+2} = \frac{5}{7}$

$P\{\text{Second event does not happen}\} = \frac{5}{5+6} = \frac{5}{11}$

$\therefore$ $P\{\text{Both the events fail to happen}\} = \frac{5}{7} \times \frac{5}{11} = \frac{25}{77}$

Therefore, the probability that at least one of the events will happen is

$$1 - P(\text{none of two happens}) = 1 - \frac{25}{77} = \frac{52}{77}$$

ILLUSTRATION 14.20

If four whole numbers taken at random are multiplied together, then find the probability that the last digit in the product is 1, 3, 7 or 9.

Sol. There are 10 digits 0, 1, 2, ..., 9 and any of which can occur in any number at the last place, i.e., at the unit place. It is obvious that if the last digit in any of the four numbers is 0, 2, 4, 5, 6, 8, then the product of any of such four numbers will not give a number having its last digit as 1, 3, 7, 9. Hence, it is necessary that the last digit in each of the four numbers must be any of the four digits 1, 3, 7, 9.

Thus, for each of the four numbers, the number of ways of selection of the last digit is 10. Favorable number of ways of selection of the last digit is 4.

Therefore, the probability that the last digit be any of the four numbers 1, 3, 7, 9 is $4/10 = 2/5$.

Hence, the required probability that the last digit in each of the four numbers is 1, 3, 7, 9 so that the last digit in their product is 1, 3, 7, 9 is $(2/5)^4 = 16/625$.

ILLUSTRATION 14.21

If A and B are two independent events, the probability that both A and B occur is 1/8 and the probability that neither of them occurs is 3/8. Find the probability of the occurrence of A.

Sol. We have,

$$P(A \cap B) = \frac{1}{8} \text{ and } P(\bar{A} \cap \bar{B}) = \frac{3}{8}$$

$$\therefore \quad P(A)P(B) = \frac{1}{8} \text{ and } P(\bar{A})P(\bar{B}) = \frac{3}{8}$$

[$\because$ A and B are independent]

Now, $P(\bar{A} \cap \bar{B}) = \frac{3}{8}$

$\Rightarrow$ $1 - P(A \cup B) = \frac{3}{8}$

or $1 - (P(A) + P(B) - P(A \cap B)) = \frac{3}{8}$

or $1 - (P(A) + P(B)) + \frac{1}{8} = \frac{3}{8}$

or $P(A) + P(B) = \frac{3}{4}$

The quadratic equation whose roots are $P(A)$ and $P(B)$ is

$$x^2 - x\{P(A) + P(B)\} + P(A)\,P(B) = 0$$

or $x^2 - \frac{3}{4}x + \frac{1}{8} = 0$

or $8x^2 - 6x + 1 = 0$

or $x = \frac{1}{2}, \frac{1}{4}$

ILLUSTRATION 14.22

The unbiased dice is rolled until a number greater than 4 appears. What is the probability that an even number of tosses is needed?

Sol. According to the question, number greater than 4 may appear in 2nd, 4th, 6th, ... trial.

Probability of getting number greater than 4 in any trial $= \frac{2}{6} = \frac{1}{3}$

$\therefore$ Probability of not getting the number greater than 4 in any trial

$$= 1 - \frac{1}{3} = \frac{2}{3}$$

Let A_i be the event that number greater than 4 occurs in the trial.

Now, result of each trial is independent of earlier trials.

So, required probability

$$= P(A_1' \cap A_2) + P(A_1' \cap A_2' \cap A_3' \cap A_4)$$
$$+ P(A_1' \cap A_2' \cap A_3' \cap A_4' \cap A_5' \cap A_6) +$$

$$= \begin{vmatrix} 1+a^2+b^2+c^2 & b^2+1 & c^2 \\ 1+a^2+b^2+c^2 & b^2 & c^2 \\ & & c^2+1 \end{vmatrix}$$

$$[C_1 \to C_1 + C_2 + C_3]$$

$$= (1+a^2+b^2+c^2)\begin{vmatrix} 1 & b^2 & c^2 \\ 1 & b^2+1 & c^2 \\ 1 & b^2 & c^2+1 \end{vmatrix}$$

$$= (1+a^2+b^2+c^2)\begin{vmatrix} 1 & b^2 & c^2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

[Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$]

$$= (1+a^2+b^2+c^2)$$

ILLUSTRATION 12.32

Prove that $\begin{vmatrix} (b+c)^2 & ba & ac \\ ba & (c+a)^2 & cb \\ ca & cb & (a+b)^2 \end{vmatrix}$

$$= \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3.$$

Sol. Multiplying R_1, R_2, R_3 by a, b, c, respectively, and dividing by abc, we get